

NEXA RETAIL STORE ANALYSIS

Q1: What is the total revenue for 2023, broken down by quarter?

Query:

```
SELECT  
  
CASE  
  
WHEN CAST(strftime('%m', order_date) AS INTEGER) BETWEEN 1 AND 3 THEN 'Q1'  
WHEN CAST(strftime('%m', order_date) AS INTEGER) BETWEEN 4 AND 6 THEN 'Q2'  
WHEN CAST(strftime('%m', order_date) AS INTEGER) BETWEEN 7 AND 9 THEN 'Q3'  
ELSE 'Q4' END AS Quarter,  
  
ROUND(SUM(total_amount),2) AS Total_Revenue  
  
FROM orders  
  
WHERE strftime('%Y',order_date)='2023'  
  
GROUP BY Quarter  
  
ORDER BY Quarter;
```

Output:

	Quarter	Total_Revenue
1	Q1	52566890.0
2	Q2	51705690.0
3	Q3	52938370.0
4	Q4	52551140.0

Execution finished without errors.

Business Interpretation:

- Revenue remained fairly consistent across all four quarters, with **Q3** generating the highest revenue (**₹52.94 million**) while **Q2** recorded the lowest (**₹51.71 million**).
- Revenue remained relatively stable across all four quarters, with the strongest quarter indicating peak sales momentum and the weakest quarter highlighting a period where demand or store performance may have softened. Management should use this quarterly trend to identify seasonal patterns and plan promotions, inventory, and staffing more effectively.

Q2: Which are the top 5 stores by total revenue? Are they all in the same city?

Query:

```
SELECT s.store_name,s.city,ROUND(SUM(o.total_amount),2) total_revenue
FROM orders o JOIN stores s ON o.store_id=s.store_id
GROUP BY s.store_id,s.store_name,s.city
ORDER BY total_revenue DESC
LIMIT 5;
```

Output:

	store_name	city	total_revenue
1	Pawar General	Nagpur	5883950.0
2	More Supermarket Viman Nagar	Surat	5753080.0
3	Jadhav Store	Nagpur	5698610.0
4	D-Mart Kothrud	Surat	5580450.0
5	Mehta Kirana	Ahmedabad	5558120.0
Execution finished without errors.			

Business Interpretation

- The highest-performing stores are spread across **Nagpur, Surat, and Ahmedabad**, indicating that strong revenue is driven by multiple regional markets rather than a single city.
- The top-performing stores reveal where NexaRetail is generating the highest value and whether performance is concentrated in one geography or spread across multiple cities. If top stores are clustered in specific locations, those markets may warrant deeper investment, while stores in other cities can be studied for operational improvements.

Q3: Which product category has the highest revenue per transaction?

Query:

```
SELECT p.category,  
ROUND(SUM(oi.line_total)*1.0/COUNT(DISTINCT oi.order_id),2) revenue_per_transaction  
FROM order_items oi  
JOIN products p ON oi.product_id=p.product_id  
GROUP BY p.category  
ORDER BY revenue_per_transaction DESC  
LIMIT 1;
```

Output:

	category	revenue_per_transaction
1	Snacks	7443.82

Execution finished without errors.

Business Interpretation:

- The Snacks category generates the highest revenue per transactions, suggesting it contributes strongly to basket value and should remain in focus for promotions and merchandising
- The category with the highest revenue per transaction contributes the most value to each customer purchase and should be prioritized in promotions, shelf placement, and bundled offers. Strengthening visibility of this category can help increase basket value and improve overall revenue efficiency.

Q4: How many stores met their monthly target in at least 8 of 12 months?

Query:

```
WITH monthly_sales AS (  
  SELECT  
    mt.store_id,  
    mt.month,  
    mt.target_amount,  
    COALESCE(SUM(o.total_amount), 0) AS actual_revenue  
  
  FROM monthly_targets mt  
  
  LEFT JOIN orders o  
    ON mt.store_id = o.store_id  
    AND strftime('%Y-%m', o.order_date) = mt.month  
  GROUP BY mt.store_id, mt.month, mt.target_amount  
)  
store_target_summary AS (  
  SELECT  
    store_id,  
  
    SUM(CASE WHEN actual_revenue >= target_amount THEN 1 ELSE 0 END) AS months_target_met  
  FROM monthly_sales  
  GROUP BY store_id  
)  
  
SELECT COUNT(*) AS stores_meeting_target_8_plus_months  
FROM store_target_summary  
WHERE months_target_met >= 8;
```

Output:

stores_meeting_target_8_plus_months	
1	3

Execution finished without errors.

Business Interpretation:

- This helps identify consistently high-performing stores. Management should study these stores' practices—such as product mix, customer volume, and local execution—and replicate them in lower-performing locations.
- Only 3 stores consistently achieved their targets for at least eight months, highlighting an opportunity to investigate what these stores do differently and replicate those practices elsewhere.

Q5: What is the month-over-month revenue growth rate? Which month had the sharpest drop?

Query:

```
WITH monthly_revenue AS (  
  SELECT  
    strftime('%Y-%m',order_date) AS month,  
    SUM(total_amount) AS revenue  
  FROM orders  
  GROUP BY month  
)  
SELECT  
  month,
```

```

revenue,

ROUND(

(revenue-LAG(revenue)

OVER(ORDER BY month))

*100.0/

LAG(revenue)

OVER(ORDER BY month),2

) AS growth_rate

FROM monthly_revenue;

```

Output:

	month	revenue	growth_rate	
1	2023-01	17700980.0	NULL	
2	2023-02	18811140.0	6.27	
3	2023-03	16054770.0	-14.65	
4	2023-04	16581560.0	3.28	
5	2023-05	17398220.0	4.93	
6	2023-06	17725910.0	1.88	
7	2023-07	17914740.0	1.07	

Execution finished without errors.

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2023-03	16054770.0	-14.65
2023-04	16581560.0	3.28
2023-05	17398220.0	4.93
2023-06	17725910.0	1.88
2023-07	17914740.0	1.07
2023-08	16402730.0	-8.44
2023-09	18620900.0	13.52
2023-10	16972560.0	-8.85
2023-11	18404210.0	8.44
2023-12	17174370.0	-6.68

Business Interpretation:

- Revenue experienced its steepest month-over-month decline in **March 2023**, indicating a significant slowdown that warrants investigation into seasonal trends or operational issues.
- Month-over-month analysis shows **sales momentum and volatility**. The sharpest drop month should be investigated for causes such as seasonality, stockouts, weak promotions, or store-level performance issues.

Q6: Which city has the highest average basket size? Which has the lowest?

Query:

```
SELECT s.city,ROUND(AVG(o.total_amount),2) average_basket_size
FROM orders o JOIN stores s ON o.store_id=s.store_id
GROUP BY s.city
ORDER BY average_basket_size DESC;
```

Output:

city	average_basket_size
Surat	15117.87
Nagpur	14834.78
Pune	14785.57
Nashik	14728.52
Aurangabad	14653.05
Ahmedabad	14595.53
Vadodara	14401.79
Solapur	14329.34
Kolhapur	14319.8
Rajkot	13998.95

Business Interpretation:

- Customers in Surat spend the most per order on average, while Rajkot has the smallest basket size, making it a potential focus area for cross-selling and upselling initiatives.
- Cities with a high basket size indicate stronger customer spend per visit and may offer best practices for product bundling and upselling. Cities with a low basket size should be targeted for cross-sell offers and assortment optimization.

Q7: Are Kirana stores or Supermarkets generating more revenue per store?

Query:

```
SELECT
    s.store_type,
    ROUND(
        SUM(o.total_amount)/
        COUNT(DISTINCT s.store_id),2
    ) AS revenue_per_store
FROM orders o
JOIN stores s
ON o.store_id=s.store_id
GROUP BY s.store_type;
```

Output:

	store_type	revenue_per_store
1	Kirana	5234148.33
2	Supermarket	5273764.0

Execution finished without errors.

Business Interpretation:

- This comparison helps determine which store format is more efficient on a per-store basis. If one format consistently outperforms the other, NexaRetail can prioritize expansion, inventory allocation, and marketing investments toward that format.
- **Supermarkets slightly outperform Kirana stores** in revenue per store, although the difference is modest, indicating both formats contribute strongly to overall revenue.

Q8: Which 3 products have declining monthly sales volume?

Query:

```
WITH monthly_product_sales AS (  
    SELECT  
        p.product_id,  
        p.product_name,  
        strftime('%Y-%m', o.order_date) AS month,  
        SUM(oi.quantity) AS total_units  
    FROM order_items oi  
    JOIN orders o  
        ON oi.order_id = o.order_id  
    JOIN products p  
        ON oi.product_id = p.product_id  
    GROUP BY p.product_id, p.product_name, strftime('%Y-%m', o.order_date)  
)  
sales_trend AS (  
    SELECT  
        product_id,  
        product_name,  
        month,  
        total_units,
```

```

        LAG(total_units) OVER (PARTITION BY product_id ORDER BY month) AS prev_units
    FROM monthly_product_sales
),
decline_count AS (
    SELECT
        product_id,
        product_name,
        SUM(CASE WHEN prev_units IS NOT NULL AND total_units < prev_units THEN 1 ELSE 0 END) AS
decline_months
    FROM sales_trend
    GROUP BY product_id, product_name
)
SELECT
    product_name,
    decline_months
FROM decline_count
ORDER BY decline_months DESC, product_name
LIMIT 3;

```

Output:

	product_name	decline_months
1	NexaGold Sunflower Oil 5L	8
2	NexaCrunch Mixture 200g	7
3	NexaGold Moong Dal 1kg	7

Execution finished without errors.

Business Interpretation:

- These products show declining unit sales over time and should be reviewed for pricing, demand shifts, or promotional support.
- Products with repeated declines in monthly unit sales should be reviewed for **pricing, visibility, competition, seasonality, or changing demand**. These items may need promotional support or SKU rationalization.

Q9: What percentage of total revenue comes from the top 10% of stores?

Query:

```
WITH store_revenue AS (  
  SELECT  
    store_id,  
    SUM(total_amount) AS revenue  
  FROM orders  
  GROUP BY store_id  
)
```

```
SELECT  
  ROUND(  
    (  
      SELECT SUM(revenue)  
    FROM(  
      SELECT revenue  
    FROM store_revenue  
    ORDER BY revenue DESC  
    LIMIT 4  
    )  
  ) * 100.0 /
```

```
(  
SELECT SUM(revenue)  
FROM store_revenue  
,2  
) AS percentage;
```

Output:

	percentage
1	10.92

Execution finished without errors.

Business Interpretation:

- This shows how **concentrated revenue is among the top-performing stores**. If a small number of stores drive a large share of revenue, the business has concentration risk and should diversify growth across more locations.
- The top **10% of stores contribute nearly 11% of total revenue**, showing that revenue is relatively well distributed rather than being concentrated in only a few locations.

Q10: If NexaRetail discontinued its bottom 3 products by revenue, how much revenue would be lost as a percentage of total?

Query:

```
WITH product_revenue AS(  
SELECT  
product_id,
```

```

SUM(line_total) AS revenue
FROM order_items
GROUP BY product_id
)
SELECT
ROUND(
(
SELECT SUM(revenue)
FROM(
SELECT revenue
FROM product_revenue
ORDER BY revenue
LIMIT 3
)
)*100.0/
(
SELECT SUM(revenue)
FROM product_revenue
),2
) AS revenue_loss_percentage;

```

Output:

	revenue_loss_percentage
1	1.92

Execution finished without errors.

Business Interpretation:

- If the bottom 3 products contribute only a very small share of total revenue, NexaRetail can consider **discontinuing them to simplify inventory and reduce operational complexity**, provided they are not strategically important for customer retention.
- Discontinuing the **three lowest-performing products would reduce total revenue by only 1.92%**, suggesting they could be candidates for rationalization if inventory or operational costs outweigh their strategic value.

